EQUINIX

March 15, 2025

Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue Alexandria, VA 22314
Via email: [redacted]

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

To Whom It May Concern:

Equinix appreciates the opportunity to provide comments on the Request for Information (RFI) on the AI Action Plan issued by the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation on behalf of the White House Office of Science and Technology Policy (OSTP) on February 6, 2025.

Equinix Introduction

Equinix is the world's digital infrastructure company and one of the largest data center operators globally, with a fleet of nearly 270 data centers in 73 metros across 34 countries, with more under development. We serve more than 10,000 enterprise, government, network and cloud customers, including 90% of the Fortune 100 and nearly 50% of the Global 2000. We opened our first data center in Ashburn, VA in 1998, and our first data centers were critical to helping the early internet scale. Equinix has since played pivotal roles in the digital transformation of enterprises and governments, provided the digital infrastructure that underpins the most critical financial ecosystems globally (including NASDAQ), and we are now positioned to provide the essential digital infrastructure that will underpin the AI revolution.

Equinix operates two types of data centers, colocation and "xScale":

- A colocation data center ("colo") is a facility where businesses and other organizations can rent space for servers and other computing hardware. Instead of maintaining their own data center, enterprises can house their IT infrastructure in a shared, secure facility that provides physical space (usually in racks or cages), power supply and backup power, cooling systems, network connectivity, physical security and technical support. Colocation data centers also facilitate interconnection (direct one-to-one connection via fiber optic cables), which provides for high speed, private data transfer from one company or enterprise to another as well as peering among network service providers. Interconnections also facilitate hybrid multi-cloud deployments and are utilized by our enterprise customers to connect to the more than 3,000 cloud and IT service providers within our facilities. Equinix boasts more than 482,000 discrete interconnections across our platform. Densely interconnected colocation data centers, such as the typical Equinix data center, create robust IT ecosystems, which facilitate further innovation for workloads that require lower latency and faster, more voluminous data transfer. This is of critical importance to AI inference. Colocation facilities represent the large majority of Equinix's current fleet.

- xScale is Equinix's hyperscale solution. Equinix may have hundreds of customers in a single colocation facility with thousands of interconnections between them, while an xScale facility may only have one or a handful of customers leasing tens or even hundreds of megawatts of capacity. Cloud service providers represent the typical xScale customer. While colocation facilities are ideal for AI inference, an xScale facility can be particularly well suited for AI model training.

We believe our unique perspective as the industry leader in colocation, interconnection, and cloud on ramps can provide valuable insight into the challenges and opportunities faced by data center operators as they work to meet AI driven demand in the U.S.

The Opportunity

As the United States stands at the forefront of an AI-driven economic and technological revolution, the Trump Administration has a unique opportunity to cement American dominance in this groundbreaking field. Global leadership, national security, and economic prosperity belongs to those who control AI infrastructure, and Equinix has built the backbone that makes it all possible. Our global network of interconnected data centers is the world's most secure, scalable, and efficient platform for AI compute, where the global technology leaders deploy their artificial intelligence.

We don't just host servers; we orchestrate the flow of AI at scale. With industry-leading, low-latency, high-performance data processing closer to end users and devices, and partnerships spanning Microsoft, Google, Amazon, Nvidia, Dell and beyond, Equinix is the neutral ground where AI's raw power meets real-world impact. Our globally interconnected footprint of data centers is not a commodity—it is a strategic asset, purpose-built to handle the exascale demands of AI workloads, from training to inference, across every industry and includes defense, intelligence community, and civilian government customers.

The following outlines why we believe Equinix is critically important to America's AI ambitions and offers strategic recommendations for the Trump Administration's forthcoming AI policy. We urge a focus on three interlocking priorities—power, supply chain, and technology talent—that will ensure the U.S. retains its dominant position in AI infrastructure. By partnering with industry leaders like Equinix, the Trump Administration can unlock the full potential of its AI strategy while advancing an "America First" agenda.

Why Equinix is Critically Important

Equinix is not just a data center provider; we are the backbone of the digital economy. Our differentiation is enabling interconnected data to unlock opportunity, and our value lies in three key areas that directly support AI development and deployment:

1. **Unmatched Infrastructure for AI Workloads**
   AI requires massive computational power, vast data storage, and low-latency connectivity—capabilities that Equinix delivers securely and at scale. Our globally interconnected footprint of International Business Exchange (IBX) data centers hosts the ecosystems where the industry's leading brands, cloud providers, and AI leaders like Microsoft, Google, AWS and NVIDIA converge. With over 10,000 customers and 2,000+ network connections, Equinix facilitates the high-speed, secure data exchange essential for training and deploying AI models. In the U.S. alone, our facilities support the critical infrastructure that powers everything from generative AI to financial markets to critical national security applications. The race for AI dominance isn't actually about who builds the best model—it's about who deploys it fastest, most reliably, and at the edge of every market near end-users. That's Equinix. We don't compete with the hyperscalers; we enable them—and our ten thousand plus customers—to win.

2. **A Strategic Asset for National Security**
   Securing U.S. dominance in AI is a national and economic security imperative, and Equinix's infrastructure is a linchpin in this effort. Our data centers are trusted by government agencies, including those responsible for national security, and defense contractors to host the most sensitive workloads, ensuring data sovereignty and resilience against cyber threats. Our global footprint, paired with a commitment to U.S.-based operations, positions us to support the Administration's goals of securing AI supply chains and maintaining technological dominance over adversaries like China.

3. **A Catalyst for Economic Growth**
   Equinix drives job creation, innovation, and investment across the country. Our U.S. operations employ thousands directly and support tens of thousands of jobs indirectly through our customers and partners. By providing the physical and digital foundation for AI, we enable businesses to scale, compete, and lead—amplifying the economic benefits of AI for American workers and the communities we operate in.

Our expertise, unique profile, and scale can make us an indispensable partner for the Trump Administration as it crafts an AI strategy to lead the world. We stand ready to be a resource.

Policy Recommendations

To maximize America's AI potential, Equinix recommends that the Trump Administration prioritize three critical areas: power, supply chain, and workforce. These align with the needs of the broader data center industry and the Trump Administration's "America First" vision.

1. **Power: Ensuring Abundant, Reliable Energy for AI Growth**

   The Challenge: Access to abundant, reliable, 24/7 baseload power is a primary consideration for data center operators, including Equinix, as they do site selection and work to meet the immense demands of AI. Power availability constraints across the U.S., driven by both generation and transmission shortfalls, are hampering data center growth today and without remediation this will only get worse.

   It is no secret that AI is highly energy intensive. A single AI training run can consume as much electricity as thousands of households, and demand is surging—with projected increases of as much as 165% by 2030. A typical, non-AI deployment in an Equinix data center might require between 5 and 10 kVA per rack while a cutting-edge AI deployment in the same rack can demand upwards of 130 kVA. This represents an increase in power requirements and power density of 13-26 times. While GPUs are projected to become more efficient, customers are expected to deploy more of them in the same rack which would negate many of the forthcoming efficiency gains. This is requiring data center operators like Equinix to rethink the way we design, cool, and power our facilities

   These realities are why Equinix has adopted an "all of the above" power strategy that includes advanced nuclear, gas fuel cells and turbines, and renewables. No single energy source will be able to meet the immense demands of AI model training or AI inference in the near term. Advanced nuclear is not ready yet, supply chain and infrastructure constraints limit the use of gas as an onsite power generation technology, and intermittent renewables are not a good match for the 24/7 demand of AI. Similarly, Equinix and the broader industry require flexibility in power deployment. This means leveraging grid power, co-located or behind the meter arrangements, and on-site power generation. It will truly take all sources in all configurations to meet AI demand.

   Additionally, Equinix works tirelessly to improve the efficiency of our operations. Improving efficiency doesn't just make business sense, it is also increasingly necessary due to the power availability constraints present across the U.S. Equinix has been at the forefront of this relentless drive towards efficiency, including becoming the first colocation provider to adopt ASHRAE A1A standards that will be raising the temperatures in our data centers and thereby lowering electricity consumption required for cooling. Equinix has also leveraged AI to improve power efficiency. We have demonstrated efficiency gains of 9% by utilizing AI controlled cooling system software in a Frankfurt data center, for instance.[1] We and the industry continue to invest in and explore innovative efficiency solutions, including liquid cooling.[2] We plan to support direct to chip liquid cooling in more than 100 of our data centers globally.[3] Equinix's commitment to driving efficiency across our fleet has led to significant reductions in our Power Usage Effectiveness ("PUE") measurement every year. PUE is a representation of how much power is used by IT equipment, such as servers, verses building operations including cooling. 1.0 is considered perfect. The Uptime Institute found that the industry average PUE in 2024 is 1.56.[4] Equinix targets a PUE of 1.3 or better for all new build facilities and has a fleetwide average of better than 1.42, down from 1.54 in 2019.

The confluence of "electrify everything", near-shoring of advanced manufacturing, AI, decades of under-investment in the grid, and an antiquated permitting system that slows the build out of new transmission and generation capacity are driving power availability constraints across the U.S., which threaten to undermine American AI dominance unless remediated. The Trump Administration has a historic opportunity to solve for these constraints and unleash energy dominance.

The Opportunity: The Trump Administration can unleash AI innovation by prioritizing energy abundance. Equinix applauds the Trump Administration's commitment to securing AI leadership and unleashing American energy. Solving for power availability constraints in the U.S. is by far the most impactful lever the Trump Administration can pull to ensure the U.S. maintains dominance in AI. Doing so will benefit all industries, including AI and advanced manufacturing.

---

[1] https://www.datacenterdynamics.com/en/news/equinix-uses-ai-to-increase-energy-efficiency-at-frankfurt-dc-by-9/

[2] https://deploy.equinix.com/blog/whats-old-is-cool-again-liquid-cooling-unlocks-data-center-sustainability/?gad_source=1&gclid=Cj0KCQjw1Yy5BhD-ARIsAI0RbXbzV1HaC1CejrcKFCsu1ZLoL9czl5oXqa6FPv3Sw_J0w_BsgleNdzYaAgCzEALw_wcB

[3] https://www.equinix.com/newsroom/press-releases/2023/12/equinix-to-accelerate-and-simplify-liquid-cooling-deployments-to-power-enterprise-ai-workloads#:~:text=(Nasdaq%3A%20EQIX)%2C%20the,45%20metros%20around%20the%20world.

[4] https://uptimeinstitute.com/resources/research-and-reports/uptime-institute-global-data-center-survey-results-2024

Recommendations:

- **Streamline Permitting:** It can take up to 10 years to permit an interregional transmission line in the U.S. currently. America's adversaries face no such constraints. We recommend that the Trump Administration work to accelerate approvals for new power generation and transmission projects through leveraging existing authorities to cut red tape and by working with Congress to pass durable permitting, planning, and siting reforms.
- **Support Diverse Energy Sources & Configurations:** To meet the step change in power requirements from AI demand, we must leverage all available sources in all configurations**.** We recommend the Trump Administration backs an "all of the above" strategy that includes advanced nuclear, gas, and renewables to meet AI's baseload and peak demands, ensuring abundant, reliable, and cost-competitive power for data centers and others. Equinix has been at the forefront of sourcing diverse forms of power. In addition to our advanced nuclear commitments, Equinix was the first data center operator to deploy natural gas powered fuel cells. Equinix now has Bloom fuel cells deployed at 19 IBX data centers in 6 states[5].

  It is also important that data center operators are able to leverage grid connected power, co-located or behind the meter arrangements, and onsite generation. There is no "silver bullet" generation source or power delivery configuration in the near term.
- **Accelerate Advanced Nuclear:** Advanced nuclear offers great promise for delivering abundant, reliable, 24/7 baseload energy, though it is still years away from deployment. Equinix was the first data center operator to commit to sourcing power from advanced small modular reactors, making a prepayment to procure at least 500MW from Oklo.[6] Since this agreement was reached, many other data center operators have announced similar commitments. The Trump Administration can endeavor to speed up the safe deployment of advanced nuclear, including small modular reactors. This will require further modernizations to licensing and permitting, the establishment of a robust domestic fuel production and recycling ecosystem, the development of a large nuclear workforce, and education on the safety and merits of advanced nuclear in the communities they will be deployed in. At

[5] https://www.bloomenergy.com/news/bloom-energy-expands-data-center-power-agreement-with-equinix-surpassing-100mw/

[6] https://www.datacenterfrontier.com/energy/article/55018022/equinix-puts-down-25m-in-data-center-nuclear-power-deal-with-sam-altmans-oklo

the same time, we must continue to invest in the existing nuclear fleet, to include uprates and restarts where possible.

An all of the above, power-first strategy will ensure Equinix and the broader data center industry can step up to meet AI driven demand, ensuring AI breakthroughs happen here and stay here while creating American jobs in energy and construction. Such a strategy will benefit all industries and ratepayers.

2. **Supply Chain: Securing Resilience, Capacity, and Bolstering Domestic Manufacturing**

The Challenge: AI relies on a complex global supply chain for hardware—power infrastructure, chips, servers, and cooling systems—much of which is vulnerable to disruptions and foreign control, especially from China. Further, lead times for critical data center components including transformers, utility scale generators, turbines, and water-cooling equipment can be years long due to limited production capacity domestically and abroad. For example, lead times for transformers have increased from approximately 50 weeks in 2021 to 120 weeks in 2024.[7]

The Opportunity: The Trump Administration's reshoring focus can strengthen domestic supply chains, aligning with Equinix's U.S.-centric operations and the industry's need for supply chain resilience and capacity.

Recommendations:

- **Bolster Domestic Manufacturing:** Expand incentives for producing AI-critical components in the U.S. and incentivize increasing production capacity in the U.S., reducing reliance on foreign suppliers and enhancing supply chain security. Consider leveraging existing authorities and working with Congress to bolster domestic production capacity of critical components.
- **Fortify Partnerships:** Encourage collaboration between data center operators like Equinix and American manufacturers to localize supply chains.
- **Targeted Trade Policies:** Use trade tools strategically to protect U.S. interests without choking innovation or access to critical materials before there is a domestic source or sufficient supply, ensuring US companies can access critical components and cutting-edge tech efficiently and without disruption.

[7] https://www.woodmac.com/news/opinion/supply-shortages-and-an-inflexible-market-give-rise-to-high-power-transformer-lead-times/

A fortified supply chain will empower the data center industry to deliver uninterrupted AI infrastructure, reinforcing America's technological edge.

3. **Workforce: Building an AI-Ready American Labor Force**

The Challenge: AI's rise demands a skilled workforce—data center technicians, engineers, and AI specialists as well as those in the building trades —yet shortages persist, risking delays in deployment and growth. Data center operators are already encountering workforce constraints in numerous markets, particularly those that have large deployments of data centers. Equinix partners with local community and technical colleges in the communities we operate, such as the Northern Virginia Community College, to design curriculum and offer scholarships, mentorship opportunities and paid internships. [8] Equinix also works to recruit veterans through partnerships with DoD's SkillBridge Program and our own Pathways program. Equinix has been recognized as a gold level employer by Military Friendly for two years running. [9]

Data center operators are also impacted by workforce shortages in construction trades that are inhibiting growth. Large scale data centers, like an Equinix xScale or other hyperscale facility, can require thousands of trades people to build over a period of years. This can strain existing workforces in a locality if there are numerous large-scale projects under development at the same time.

The Opportunity: The Trump Administration can harness America's entrepreneurial spirit to upskill workers, aligning with the industry's need for talent and further bolstering positive economic impact for the communities we build and operate in.

Recommendations:

- **Expand Training Programs:** Fund career and technical education and apprenticeship initiatives in partnership with industry, leveraging Equinix's expertise to train workers for data center and AI roles. Similarly, support the robust growth of the talent pipeline for the building trades who construct AI data centers.
- **Promote STEM Education:** Invest in K-12 and higher education to build a strong pipeline of American talent, ensuring US companies have the workforce to scale.

---

[8] https://www.nvcc.edu/about/news/press-releases/2021/Equinix-scholarship.html
[9] https://blog.equinix.com/blog/2023/11/10/bridging-service-to-tech-we-support-veteran-career-growth/

- **Support Reskilling:** Back programs to transition workers from declining industries and veterans into AI-related fields. A robust workforce will fuel the industry's expansion and cement U.S. leadership in AI innovation.

Conclusion

Equinix stands ready to be a strategic partner in the Trump Administration's AI strategy. Our unmatched globally interconnected infrastructure, economic impact, and national and economic security contributions make us a defining player in this space. By focusing on power, supply chain, and workforce, the Trump Administration can secure U.S. leadership in AI while advancing policies that benefit all industries.

We look forward to continued engagement with, and serving as a resource for, the Trump Administration as it shapes it's AI strategy. Together, we can remove barriers, drive innovation, and ensure America continues to lead the world in artificial intelligence.

Submitted by:

David Young
Head of U.S. Federal & Americas Public Policy and Government Affairs
Equinix, Inc.